Fiscal Year:	FY 2026	**Task Last Updated:**	FY 06/24/2026
PI Name:	Sundaresan, Alamelu Ph.D.		
Project Title:	Antibody Somatic Hypermutation Under Spaceflight Conditions		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Cell Culture		
PI Email:	alamelu.sundaresan@tsu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	(713) 313-7926
Organization Name:	Texas Southern University		
PI Address 1:	New Science Building, Rm 203-M		
PI Address 2:	3100 Cleburne Street		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77004	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 ILSRA--Flight Opportunities for Space Life Sciences (non-US proposers)
Start Date:	09/01/2020	**End Date:**	08/31/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
Contact Email:	harry.jones@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 08/31/2026 per PI and NSSC information (Ed., 9/8/25) NOTE: End date changed to 08/31/2025 per F. Hernandez/ARC (Ed., 8/29/24) NOTE: End date changed to 08/31/2024 per PI and NSSC information (Ed., 9/2/23)		
Key Personnel Changes/Previous PI:	None.		
COI Name (Institution):	Frippiat, Jean-Pol Ph.D. (European Space Agency (ESA) PI: Lorraine University, France)		
Grant/Contract No.:	80NSSC20K1308		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Dr Alamelu Sundaresan is the U.S. Co-Investigator on this European Space Agency (ESA)-sponsored project. ESA/CNES (Centre National d'Etudes Spatiales/ French National Centre for Space Studies) Principal Investigator is Jean-Pol Frippiat, Ph.D., EA7300, Lorraine-University, France. Immunity is a key biological system affected by spaceflight. Our previous space experiment suggested that spaceflight conditions decrease the frequency of antibody somatic hypermutations (Bascove et al., 2011, FASEB Journal). This observation could imply less efficient maturation of antibody affinity in space, which would in turn decrease the efficiency of the immune response. Her part includes all analog spaceflight studies and post flight signal transduction studies. Because these data were obtained using a small number of amphibians (n=2), we would like to 1) confirm these results using a constitutively hypermutating human B cell line and 2) understand how the genetic machinery that introduces these mutations is altered in space. To reach these goals, we plan to culture a constitutively hypermutating human B cell line in the International Space Station (ISS). Cells will be grown onboard the ISS and samples of this culture will be fixed in-flight to determine spaceflight-induced changes. The same experiment will be performed on Earth as a control. Space samples will be returned, when possible, to laboratories where they and control samples will be used for 1) somatic hypermutation analyses to determine if their frequency and pattern along an immunoglobulin VDJ (variability, diversity, and joining) rearrangement are modified and 2) gene expression profiling using microarrays to understand how the somatic hypermutation machinery that introduces these mutations is affected by spaceflight conditions. These analyses will expand our knowledge regarding the effects of spaceflight conditions on antibody somatic hypermutations and consequently, contribute to a better understanding of spaceflight-associated weakening of the immune system, which is an important issue for the preparation of long-term crewed space missions. Furthermore, because several studies have suggested that age-related deficiencies in the generation of high-affinity antibodies might be due to impaired somatic hypermutation, this project could also provide useful data for understanding age-related changes in the immune system. The ISS flight experiment will be manifested in future and the experiment time line is thus tentative and could be altered. Bascove M, Guéguinou N, Schaerlinger B, Gauquelin-Koch G, Frippiat JP. Decrease in antibody somatic hypermutation frequency under extreme, extended spaceflight conditions. FASEB J. 2011 Sep;25(9):2947-55. https:// . PubMed PMID: 21593434.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These analyses will improve our knowledge about the effects of spaceflight conditions on antibody somatic hypermutations and consequently contribute to a better understanding of spaceflight-associated immune system weakening, an important issue for the preparation of long-term manned space missions. Furthermore, since several studies suggested that age-related deficiency in generating high-affinity antibodies might be due to impaired somatic hypermutations, this project could also provide useful data to understand age-related changes of the immune system. At present no data is available on B cell antibody somatic hyper mutation in space or spaceflight analog human systems. Immune dysregulation is one of the most consistently reported physiological consequences of spaceflight. Astronauts exposed to long-duration missions frequently exhibit impaired immune responses, including reduced lymphocyte activation, altered cytokine production, diminished natural killer cell activity, and reactivation of latent viral infections such as Epstein–Barr virus and varicella-zoster virus (Crucian et al., 2018). Of particular concern is the impairment of adaptive immune function, which directly affects the body's ability to mount antigen-specific responses and generate immunological memory. Evidence suggests that both T-cell and B-cell compartments are affected, with reductions in proliferative capacity and altered signaling thresholds. These immune alterations have significant implications for astronaut health during long-duration missions to the Moon, Mars, and deep space, where exposure to pathogens combined with reduced immune surveillance could increase infection risk and compromise vaccine efficacy. Understanding the mechanistic basis of immune dysregulation under microgravity is therefore not only of fundamental scientific interest but also of critical importance for space medicine and the development of effective countermeasures to preserve immune function in extraterrestrial environments.
Task Progress:	Our grant performance actions will depend on the flight manifest date, which has not been decided yet. This is stated in the grant. The flight experiment might be tentatively manifested later this year, as per the European Principal Investigator (PI). However, this is not firm and could be moved or not occur. There is considerable ground-based work which needs to be completed as part of this PI's task. Two students' Ph.D. projects rely on this grant, and some very good results have been obtained so far. Hence, thrust is on ground-based analog work on somatic hypermutation (SHM) during the No-Cost Extension (NCE) period. Hence, we request NCE to complete their dissertation work. More time is needed based on analysis needed to complete the ground-based work. Their dissertations are hinging on these experiments. Optimization experiments investigating a new alternative protein marker for somatic hypermutation were initiated as part of this PI's goals. This was not known earlier and will be a good addition to the sequencing protocol followed to analyze hypermutation at present. The PI spent Summer 2025 and will spend Summer 2026 working extensively on these experiments to establish these alternatives signaling molecules as valid indicators of somatic hypermutation. This is an innovation in itself. Previously, sequencing was the only method available to determine SHM, but these alternatives offer analysis even if sample volume returned is small. SPECIFIC OBJECTIVES AND RESULTS Immune dysregulation is one of the most consistently reported physiological consequences of spaceflight. Astronauts exposed to long-duration missions frequently exhibit impaired immune responses, including reduced lymphocyte activation, altered cytokine production, diminished natural killer cell activity, and reactivation of latent viral infections such as Epstein–Barr virus and varicella-zoster virus (Crucian et al., 2018). Of particular concern is the impairment of adaptive immune function, which directly affects the body's ability to mount antigen-specific responses and generate immunological memory. Evidence suggests that both T-cell and B-cell compartments are affected, with reductions in proliferative capacity and altered signaling thresholds. These immune alterations have significant implications for

	astronaut health during long-duration missions to the Moon, Mars, and deep space, where exposure to pathogens combined with reduced immune surveillance could increase infection risk and compromise vaccine efficacy. Understanding the mechanistic basis of immune dysregulation under microgravity is therefore not only of fundamental scientific interest but also of critical importance for space medicine and the development of effective countermeasures to preserve immune function in extraterrestrial environments.
Bibliography Type:	Description: (Last Updated: 07/02/2026)
Abstracts for Journals and Proceedings	Mann V, Chikkula K, Sundaresan A. "Microgravity-induced alterations in human B cell function." 2026 International Society for Gravitational Physiology (ISGP) Conference, Cologne, Germany, May 22-29, 2026. Abstracts. 2026 International Society for Gravitational Physiology (ISGP) Conference, Cologne, Germany, May 22-29, 2026. , May-2026
Books/Book Chapters	Mann V, Sundaresan A. "Regenerative medicine and spaceflight." in "Precision Medicine for Long and Safe Permanence of Humans in Space." Ed. C. Krittanawong. San Diego, CA: Academic Press, 2025, p.163-177. http://dx.doi.org/10.1016/B978-0-443-22259-7.00039-4 , Jan-2025